GOVERNMENT OF KHYBER PAKHTUNKHWA

KHYBER PAKHTUNKHWA TEXTBOOK BOARD PESHAWAR

OFFICIAL PRESCRIBED E-CURRICULUM TEXTBOOK

MATHEMATICS CLASS 8

CLASS / GRADE: 8th (Middle)

Subject: Mathematics | Language: English

AUTHENTIC CURRICULUM SYLLABUS & CHAPTER BLUEPRINTS

· Total Complete Curriculum Chapters: 7 Units

· Designed as per National Curriculum & KPK BISEP Standards

· Contains complete theory, conceptual exercises, MCQs & solutions

· Fully synchronized with ExamForge AI Examination Paper Generator

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UNIT 1: REAL NUMBERS AND NUMBER SYSTEM

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Real Numbers and Number System.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 2: SQUARE ROOTS AND CUBE ROOTS

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Square Roots and Cube Roots.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 3: NUMBER SYSTEMS: BASE 2, BASE 5, BASE 8, BASE 10

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Number Systems: Base 2, Base 5, Base 8, Base 10.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 4: FINANCIAL ARITHMETIC: BANKING, COMPOUND INTEREST &

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Financial Arithmetic: Banking, Compound Interest & Insurance.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 5: POLYNOMIALS: DIVISION AND IDENTITIES

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Polynomials: Division and Identities.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 6: SIMULTANEOUS LINEAR EQUATIONS

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Simultaneous Linear Equations.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 7: SURFACE AREA AND VOLUME OF CYLINDERS & CONES

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Surface Area and Volume of Cylinders & Cones.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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